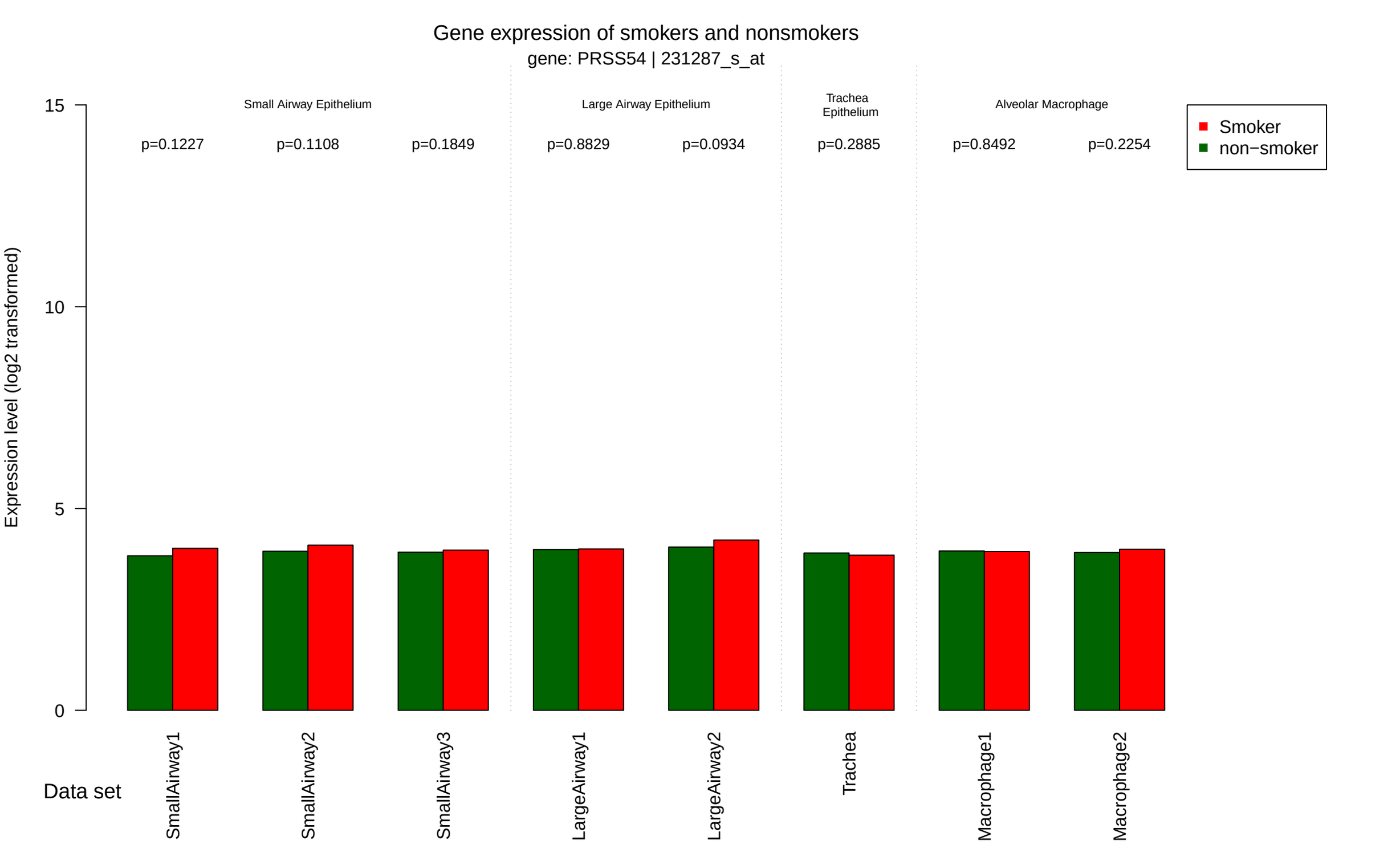
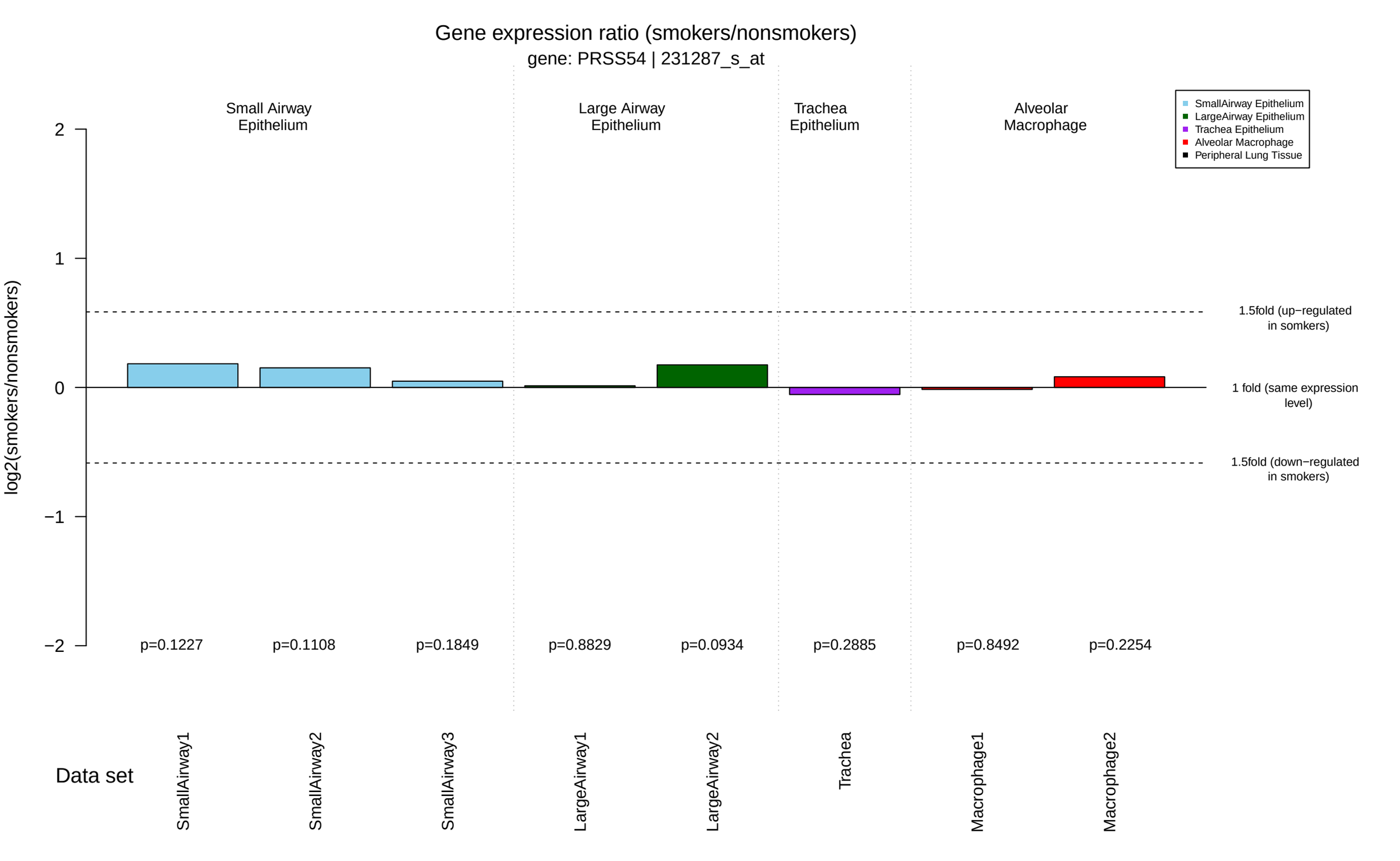
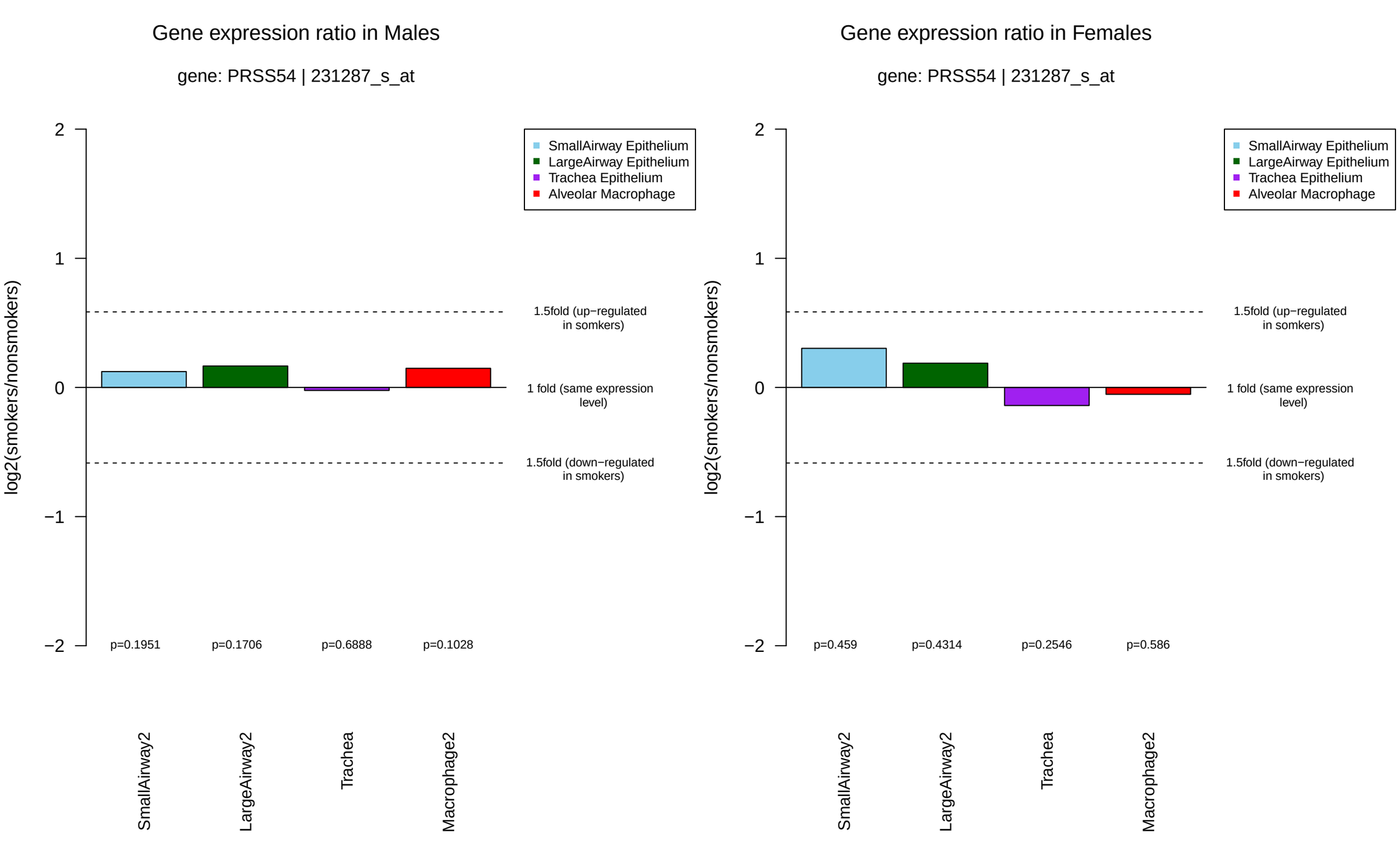
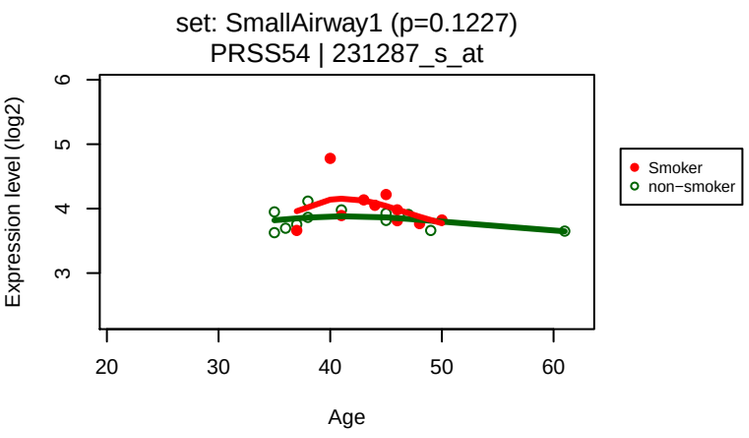
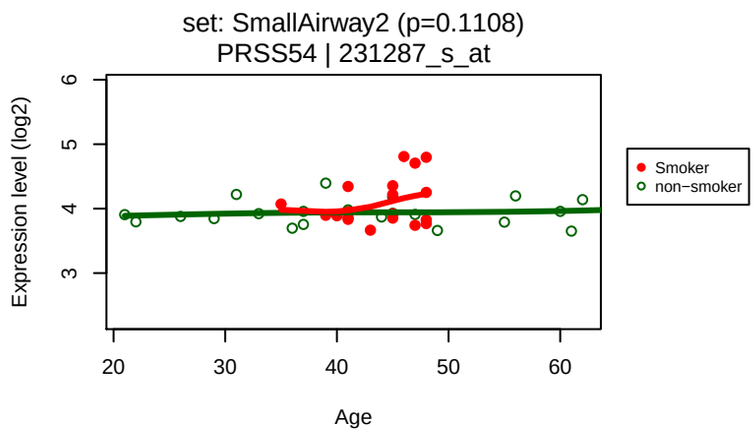
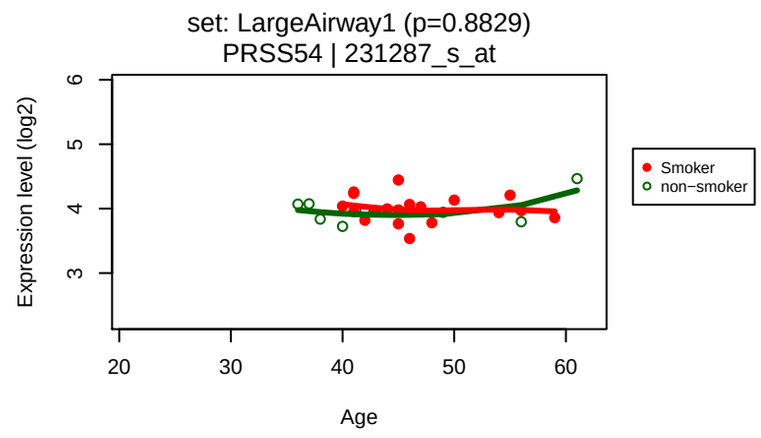

Gene expression of smokers and nonsmokers
gene: PRSS54 | 231287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1227
p=0.1108
p=0.1849
p=0.8829
p=0.0934
p=0.2885
p=0.8492
p=0.2254
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

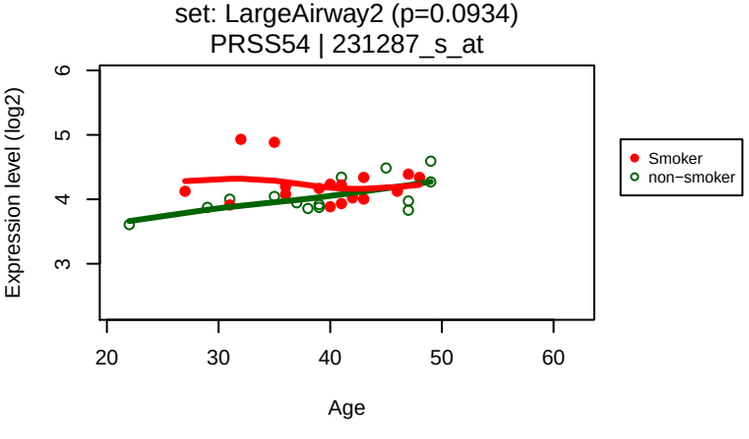

Gene expression ratio (smokers/nonsmokers)
gene: PRSS54 | 231287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1227
p=0.1108
p=0.1849
p=0.8829
p=0.0934
p=0.2885
p=0.8492
p=0.2254
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

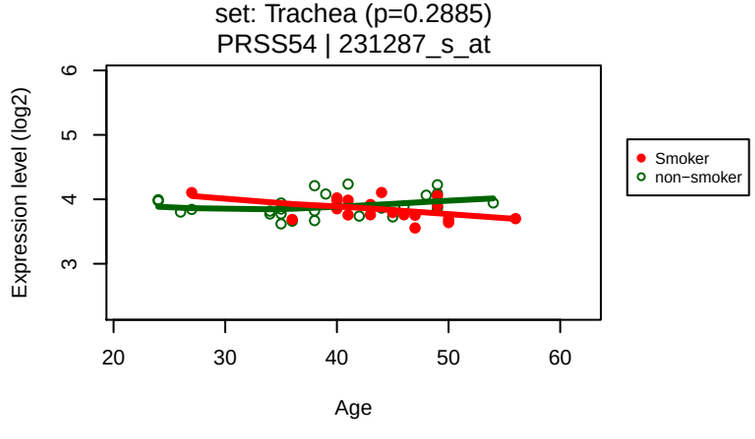

Gene expression ratio in Males
gene: PRSS54 | 231287_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1951
p=0.1706
p=0.6888
p=0.1028
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS54 | 231287_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.459
p=0.4314
p=0.2546
p=0.586
SmallAirway2
LargeAirway2
Trachea
Macrophage2

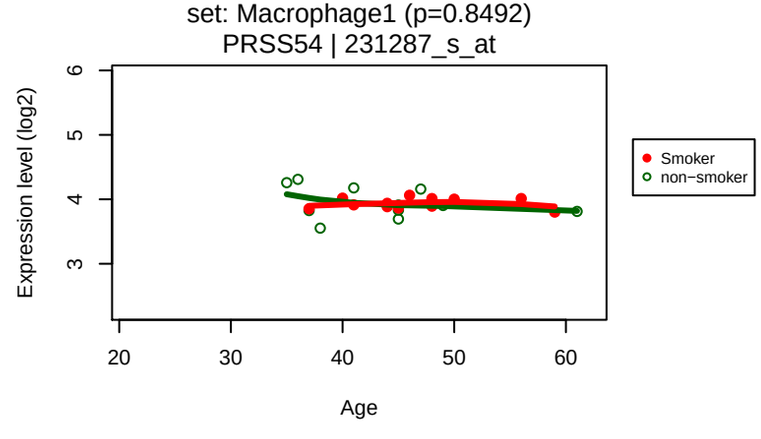

set: SmallAirway1 (p=0.1227)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker

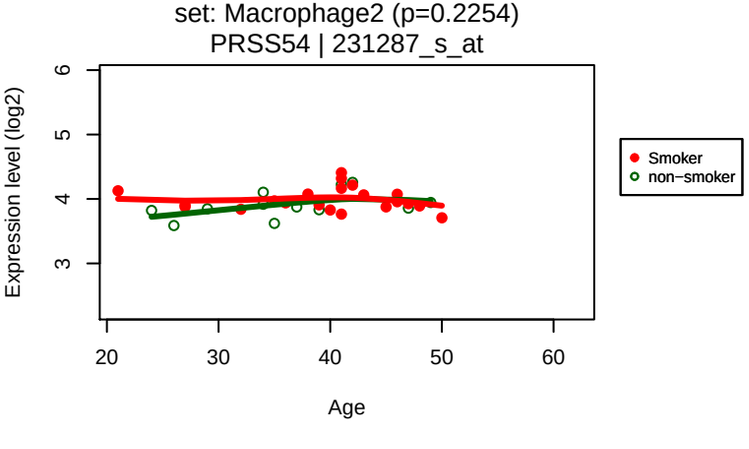

set: SmallAirway2 (p=0.1108)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker

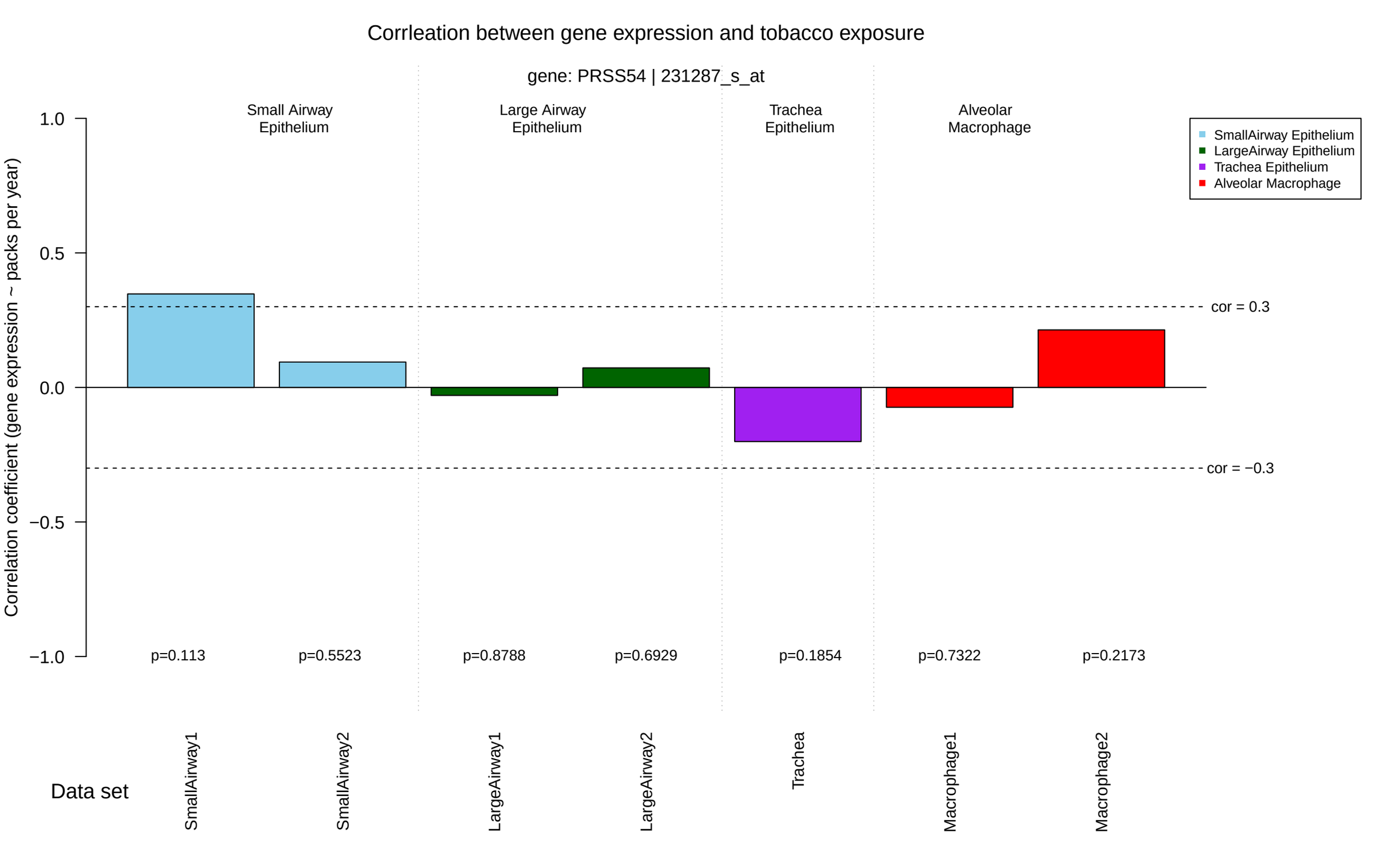

set: LargeAirway1 (p=0.8829)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0934)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2885)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8492)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2254)
PRSS54 | 231287_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS54 | 231287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.113
p=0.5523
p=0.8788
p=0.6929
p=0.1854
p=0.7322
p=0.2173
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2